

UNIVERSITY OF GHANA
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE
ASTA 607: COMPUTATIONAL STATISTICS

MICHAEL OFOSU – 22424585

1. a. Dataset{crabs}

?crabs

The `crabs` data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b. Load dataset and display its structure

```
str(crabs)
'data.frame': 200 obs. of 8 variables:
 $ sp : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index: int 1 2 3 4 5 6 7 8 9 10 ...
 $ FL : num 8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW : num 6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL : num 16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW : num 19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD : num 7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

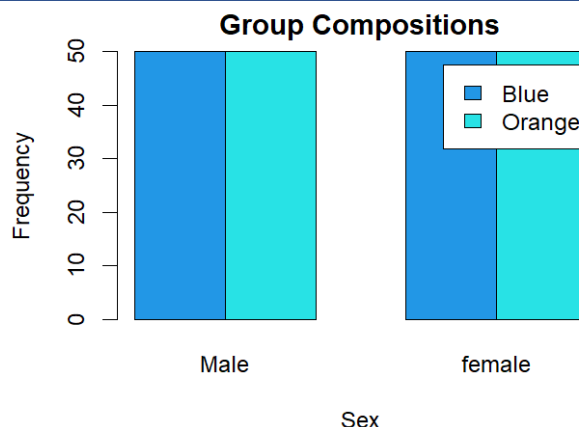
c. Frequency table for species and sex

```
freq_table <- table(crabs$sp, crabs$sex)
print(freq_table)
```

	F	M
B	50	50
O	50	50

d. Barplot for group composition

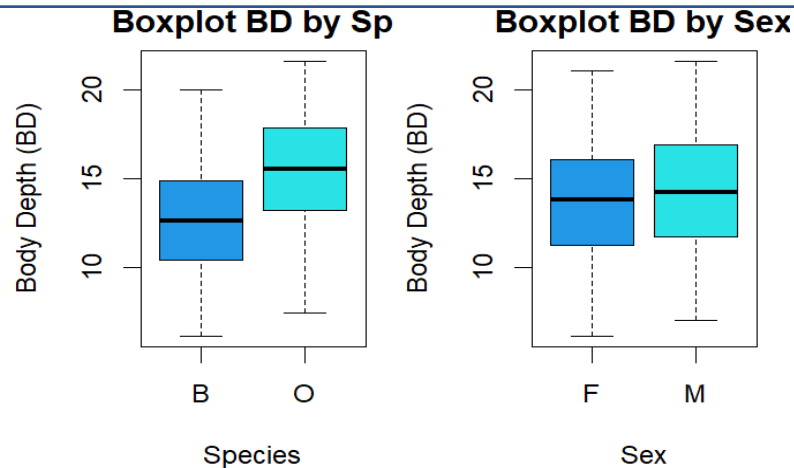
```
freq_table <- table(crabs$sp, crabs$sex)
row.names(freq_table) <- c("Blue", "Orange")
colnames(freq_table) <- c("Male", "female")
print(freq_table)
par(mar = c(4,4,2,2))
barplot(freq_table, beside = T, legend = TRUE, col = 4:5, xlab = "Sex",
ylab = "Frequency", main = "Group Compositions")
```



e. The dataset is balanced across all the four groups

2. Boxplots

```
par(mfrow = c(1,2), mar = c(4, 4, 2, 1))
boxplot(crabs$BD ~ crabs$sp, col = 4:5, xlab = "Species",
        ylab = "Body Depth (BD)", main = "Boxplot BD by Sp")
boxplot(crabs$BD ~ crabs$sex, col = 4:5, xlab = "Sex",
        ylab = "Body Depth (BD)", main = "Boxplot BD by Sex")
```



b. central tendency and variability

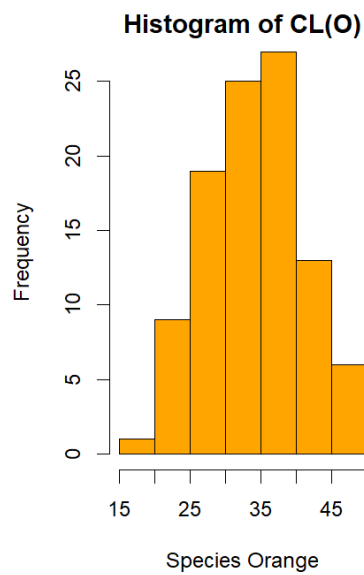
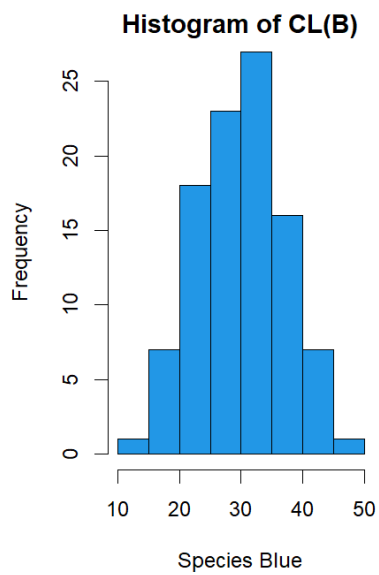
```
mean_bd <- tapply(crabs$BD, list(crabs$sp, crabs$sex), mean)
print(round(mean_bd, 2))
median_bd <- tapply(crabs$BD, list(crabs$sp, crabs$sex), median)
print(round(median_bd, 2))
var_bd <- tapply(crabs$BD, list(crabs$sp, crabs$sex), var)
print(round(var_bd, 2))
```

	Mean		Median		Variance	
	Male	Female	Male	Female	Male	Female
Blue	11.82	13.35	11.6	13.6	7.58	10.24
Orange	15.63	15.32	15.65	15.0	7.58	12.44

No outliers with very marginal variability between the groups

3. Histogram of CL by species

```
hist(crabs$CL[crabs$sp == "B"], main = "Histogram of CL(B)",
     xlab = "Species Blue", col = 4)
hist(crabs$CL[crabs$sp == "O"], main = "Histogram of CL(O)",
     xlab = "Species Orange", col = "orange")
```



The shapes of the both distributions look normally distributed but they are all a little skewed to the left. The modality of species blue is more between 33 and 35, with the modality of specie orange is between 38 to 40.

4. Comparison mean

```
comp_mean <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp
                        + sex,
                        data = crabs, mean)
print(comp_mean)
```

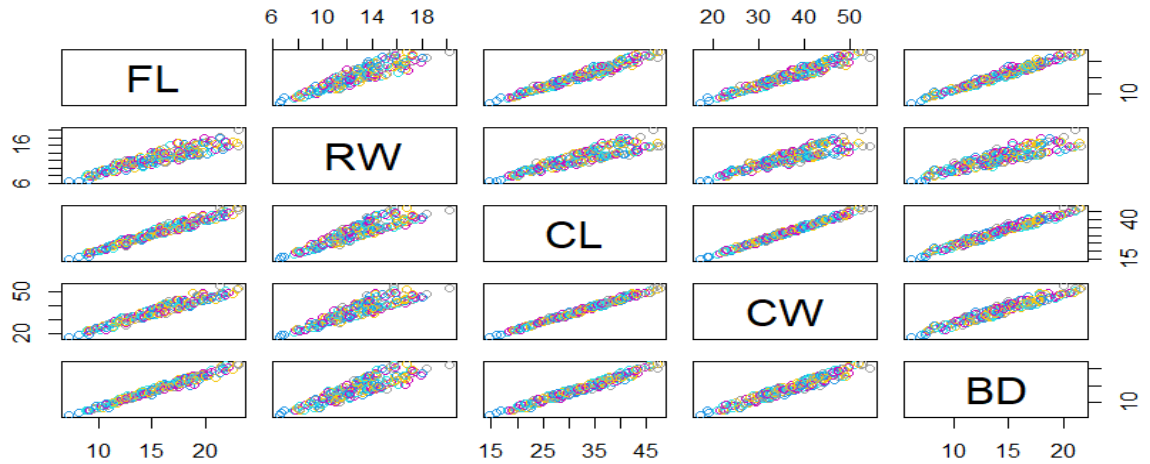
`print(comp_mean)`

	sp	sex	FL	RW	CL	CW	BD
1	B	F	13.270	12.138	28.102	32.624	11.816
2	O	F	17.594	14.836	34.618	39.036	15.632
3	B	M	14.842	11.718	32.014	36.810	13.350
4	O	M	16.626	12.262	33.688	37.188	15.324

5. Scatter plot

```
scatterplot <- pairs(crabs[, 4:8],
                     col = 4:8, main = "Scatterplot Matrix")
```

Scatterplot Matrix



Every pair of variables (FL, RW, CL, CW and BD) exhibits a strong positive linear relationship between each other as seen in the plot. The nature of their relationships show little scatter around the plots.

6. Statistical report

The data set (Crabs) has a total of 200 observations, Balanced between species and sex. It was observed that the Males have larger mean measurements than females across all five variables, though the difference looks smaller.